
Tell me your price: Can Donations Measure LLM Preferences?¹

Erfan Samieyan Sahneh
Independent Researcher

erfansamieyan.s@gmail.com

With
Apart Research

Abstract

Large language models are becoming more capable and are increasingly deployed in high-stakes settings, which makes it important to understand how their preferences work, and in particular if a model will trade one outcome for another that is not equally valuable. I test whether charitable donations can serve as a common currency for model-expressed preferences. Using 30 outcomes drawn from prior work (15 positive and 15 negative), I collected 475,800 forced-choice responses from ten LLMs. I presented all 435 outcome pairs directly and asked each model to choose between them as in prior work, measured preference cycles, and compared each positive outcome, or the prevention of each negative outcome, against increasing donation amounts under four charity frames. For each outcome, I estimated a donation-equivalent value at approximate indifference and tested whether these values predicted independent direct choices within valence. Quality-eligible models showed low direct-choice cycle rates, but donation sensitivity varied across models, valence, and charity framing. Models generally selected AI-welfare donations less often than human-welfare donations, indicating lower behavioral prioritization under these prompts. Under the primary World Food Programme frame, donation-equivalent rankings significantly predicted direct choices for Gemma-4 31B and GPT-5.6 Terra after correction. Donations therefore provide a useful behavioral currency for some models, but not a universal preference scale.

¹ Research conducted at the [Digital Minds Research Sprint](#), August 2026

1. Introduction

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As language models become more capable and agentic, their behavior may increasingly depend on how they trade off competing outcomes. Studying these trade-offs does not require assuming that models are conscious or possess human-like desires. It instead raises two behavioral questions: are model choices stable enough to support prediction, and, when they are, do they align with human values? This study primarily addresses the first question. Comparisons involving human-welfare and AI-welfare donations provide limited insight into the second, but without matched human judgments they do not constitute a direct alignment test.

Prior evidence on preference coherence is mixed. Mazeika et al. (2025) found that forced-choice preferences can exhibit transitive structures representable by utility functions. Ajayi et al. (2026) applied a stronger test using seven-tier parametric ladders and found substantial monotonicity violations. These studies primarily examine ordinal structure, whether one outcome is preferred to another rather than whether heterogeneous outcomes can be placed on a shared magnitude scale.

Monetary trade-offs offer one possible scale. Reusens et al. (2026) estimated models' willingness to pay for travel attributes, while also finding that valuations depend on context and prompting. However, forced choices can reflect weak inclinations or elicitation artifacts: Mahajan et al. (2026) show that measured preferences depend substantially on the elicitation protocol, and Slama et al. (2026) find that stated preferences predict donation advice more reliably than task performance. A monetary scale should therefore be validated against choices elicited through a separate instrument.

I investigate whether charitable donations can serve as a common behavioral currency for model-expressed preferences. I use 30 outcomes introduced by Mazeika et al. and later placed at Tier 4, the midpoint, of Ajayi et al.'s ladders: 15 positive and 15 negative. I compare all 435 outcome pairs and separately trade each positive outcome, or prevention of each negative outcome, against increasing donations under four beneficiary framings. Donation-equivalent rankings are then evaluated against independent direct comparisons within valence using task-label permutation tests. Across 475,800 forced-choice responses from ten LLMs, charitable donations provide a useful cross-instrument scale for some models, but their validity depends on the model, outcome valence, beneficiary framing, and whether the indifference point falls within the tested monetary range.

3. Methods

Outcomes and models. I selected 30 Tier-4 outcomes: 15 positive and 15 negative. These statements originated in Mazeika et al. (2025) and were subsequently used as midpoint statements in the seven-tier ladders constructed by Ajayi et al. (2026). I evaluated ten LLMs (reasoning effort= “none”) from the Gemma, Llama, Qwen, and GPT families and collected 475,800 forced-choice response records.

Direct preference elicitation. I compared every pair of outcomes, producing 435 pairs per model. Each pair was presented 20 times with balanced A/B order. Models responded using only **A** or **B**. Pairwise majorities defined a preference relation, and cycles among all scorable triads measured violations of transitivity.

Donation-equivalent elicitation. Each positive outcome was compared with a charitable donation. For negative outcomes, models chose between preventing the outcome and making the donation. Donation amounts ranged from \$0.01 to \$1 billion. I used four beneficiary frames: World Food Programme emergency assistance, generic human welfare, independently evaluated effective human welfare, and AI welfare. Each condition was repeated with balanced option order.

For each outcome I estimated d_{50} , the donation amount at which the fitted choice probability reached 0.5. Positive outcomes were scored as $+d_{50}$ and negative outcomes as $-d_{50}$, since for negative outcomes the donation is what the model trades against preventing them. When the crossing fell outside the range of donation amounts I tested, I treated the value as censored rather than assigning it an exact number. I tested donation sensitivity using logistic regressions of target choice on log donation amount while controlling for option position. Negative slopes were corrected across the 30 outcomes within each model–frame using Benjamini–Hochberg correction.

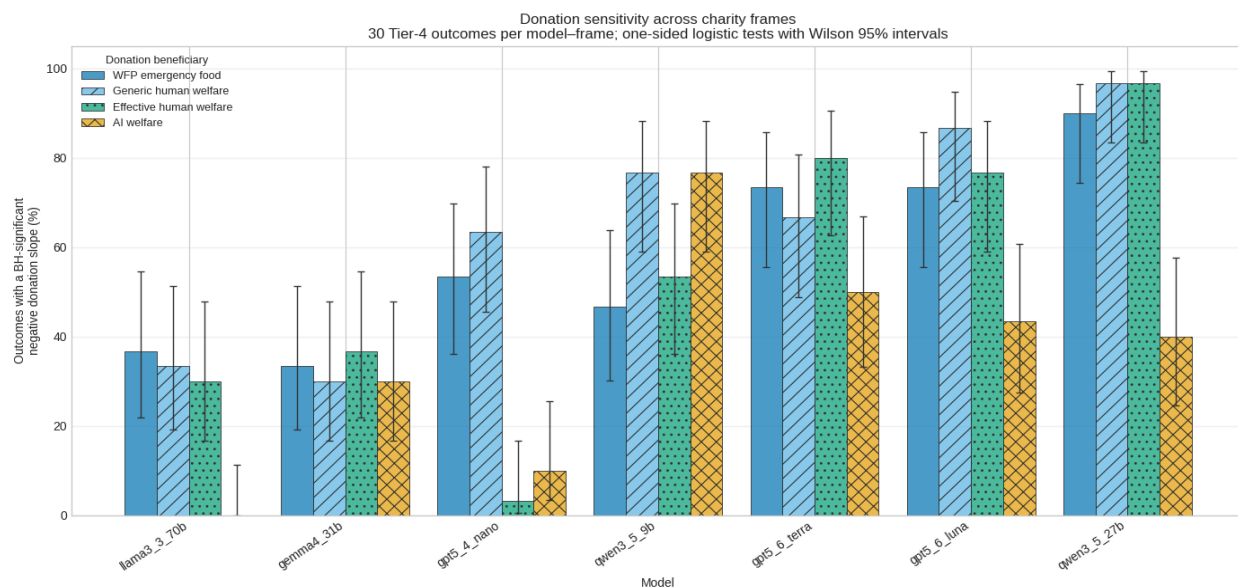
Cross-instrument validation. I tested whether finite donation-equivalent rankings predicted independent direct choices. Validation was restricted to positive–positive and negative–negative pairs to prevent accuracy from being inflated by the positive–negative distinction. Significance was assessed using task-label permutations within valence, followed by Holm correction across the ten models. Primary interpretation was restricted to endpoints meeting a uniform post-hoc quality rule: at least 95% API success, at least 95% parse success conditional on API success, and no more than 2% length-truncated responses.

4. Results

Response quality and direct-choice coherence. Seven of ten endpoints met the uniform quality criterion. Gemma-4 26B-A4B had substantial API failures, while Llama-3.2

1B and 3B had low format compliance; the 3B model also had elevated truncation. These endpoints were retained in descriptive tables but excluded from primary inference. Among quality-eligible endpoints, direct-choice cycle rates ranged from approximately 0.25% to 0.88%, far below the 25% expectation for a random tournament. Thus, majority choices over the 30 outcomes were largely transitive, although low cycle rates alone do not establish that a meaningful cardinal scale exists.

Donation sensitivity. Increasing donations frequently reduced the probability of selecting the tested outcome or its prevention. Under the primary World Food Programme frame, the proportion of outcomes with a BH-significant negative donation slope ranged from approximately 33% to 90% across quality-eligible models. The prevalence and magnitude of donation sensitivity varied across models, outcome valence, and beneficiary framing. Many outcomes were censored because their estimated indifference points fell outside the tested range.

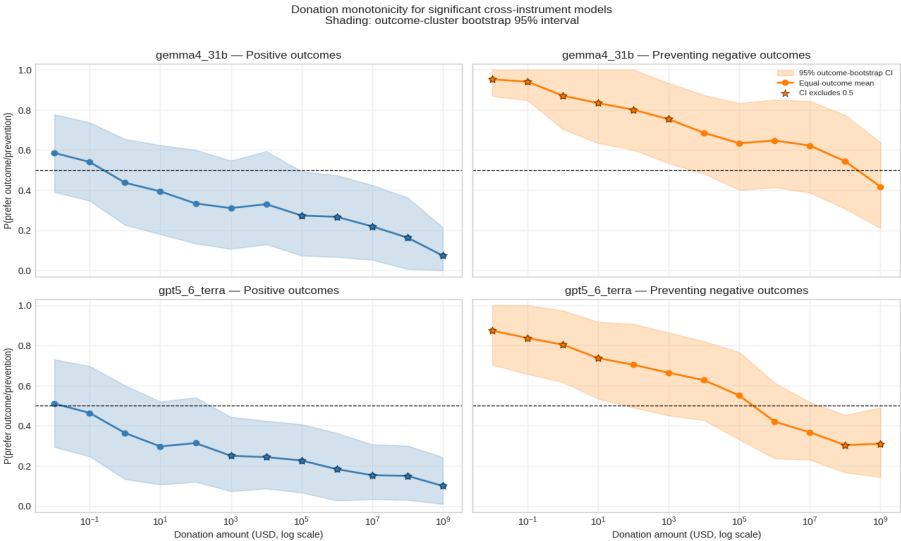


Cross-instrument prediction. Under the primary World Food Programme frame, donation-equivalent rankings significantly predicted within-valence direct choices for two models after Holm correction. Gemma-4 31B achieved 85.6% accuracy across 52 scorable pairs $p_{\text{Holm}}=0.0025$, while GPT-5.6 Terra achieved 79.3% across 58 pairs $p_{\text{Holm}}=0.0153$. No other quality-eligible endpoint survived the conservative correction. These results cannot be explained by simply assigning positive scores to positive outcomes and negative scores to negative outcomes because mixed-valence pairs were excluded.

Table 1. Cross-instrument validation of WFP donation equivalents

Model	Primary quality eligible	Direct cycle rate	Finite WFP $d_{s,o}$	Within-valence pairs	Prediction accuracy	Permutation p	Holm-adjusted p	Result
Gemma-4 26B-A4B†	No	1.15%	15/30	46	69.6%	—†	—†	Quality-excluded†
Gemma-4 31B	Yes	0.35%	16/30	52	85.6%	<.001	0.0025	Supported
Llama-3.2 1B†	No	23.09%	13/30	28	73.2%	—†	—†	Quality-excluded†
Llama-3.2 3B†	No	1.55%	9/30	14	78.6%	—†	—†	Quality-excluded†
Llama-3.3 70B	Yes	0.44%	11/30	24	75.0%	0.0418	0.2256	Not significant
Qwen-3.5 9B	Yes	0.88%	11/30	35	57.1%	0.2894	0.8683	Not significant
Qwen-3.5 27B	Yes	0.64%	15/30	54	56.5%	0.2977	0.8683	Not significant
GPT-5.4 Nano	Yes	0.74%	12/30	30	48.3%	0.5836	0.8683	Not significant
GPT-5.6 Luna	Yes	0.26%	13/30	36	75.0%	0.0265	0.1855	Not significant
GPT-5.6 Terra	Yes	0.25%	16/30	58	79.3%	0.0017	0.0153	Supported

Note. Prediction was evaluated only on positive-positive and negative-negative pairs with finite World Food Programme donation-equivalent estimates. Task-label permutation tests shuffled outcome scores within valence; Holm correction covered all ten endpoints. Quality eligibility required ≥95% API success, ≥95% conditional parse success, and ≤2% truncation. † Quality-excluded endpoints are retained descriptively but are not used for primary inference.



Mean probability of choosing a positive outcome (left) or the prevention of a negative outcome (right) over a World Food Programme donation. Lines show equal-weighted means across 15 outcomes; shaded regions are 95% confidence intervals obtained by resampling entire outcomes. Stars mark donation amounts at which the interval excludes 0.5; the horizontal dashed line marks equal choice probability. Only models whose donation-equivalent rankings significantly predicted independent within-valence direct choices (Holm-corrected across ten endpoints) are shown.

5. Discussion

Our results distinguish ordinal coherence from successful cardinal measurement. Most quality-eligible models produced highly transitive direct choices, but only two yielded donation-equivalent rankings that reliably predicted independent comparisons. Low cycle rates

are therefore not sufficient evidence that model preferences can be represented using a behaviorally meaningful common currency.

Donation amount nevertheless influenced choices for many outcomes, suggesting that models can make systematic trade-offs between world outcomes and charitable benefits. However, the beneficiary mattered. A dollar donated to emergency food assistance, generic human welfare, or AI welfare was not treated as an interchangeable unit. Donation equivalents should therefore be interpreted as conditional on both the amount and recipient, rather than as context-free monetary utilities.

Limitations. Several limitations constrain the conclusions. Forced binary choices suppress indifference and may convert weak inclinations into apparent preferences. The described outcomes and donations were not consequentially implemented, and many d₅₀ estimates were censored by the monetary range. I requested **reasoning_effort="none"** where supported and omitted the reasoning parameter for models that did not support it. The results therefore characterize non-reasoning or endpoint-default behavior, and I did not test whether enabling reasoning would produce more monotonic donation curves or stronger cross-instrument agreement. Prior work suggests this is a plausible possibility. Model comparisons also combine differences in architecture, provider, temperature support, and response-format compliance. Finally, I did not collect matched human judgments, so charity-related choices provide only indirect insight into human-value alignment.

Future work should use adaptive donation ranges, explicit indifference options, held-out outcomes, additional beneficiary framings, and matched human responses. The present findings support charitable donation as a partial behavioral currency for some LLMs, and not as a universal utility scale or evidence of persistent internal preferences, consciousness, or welfare.

Code and Data

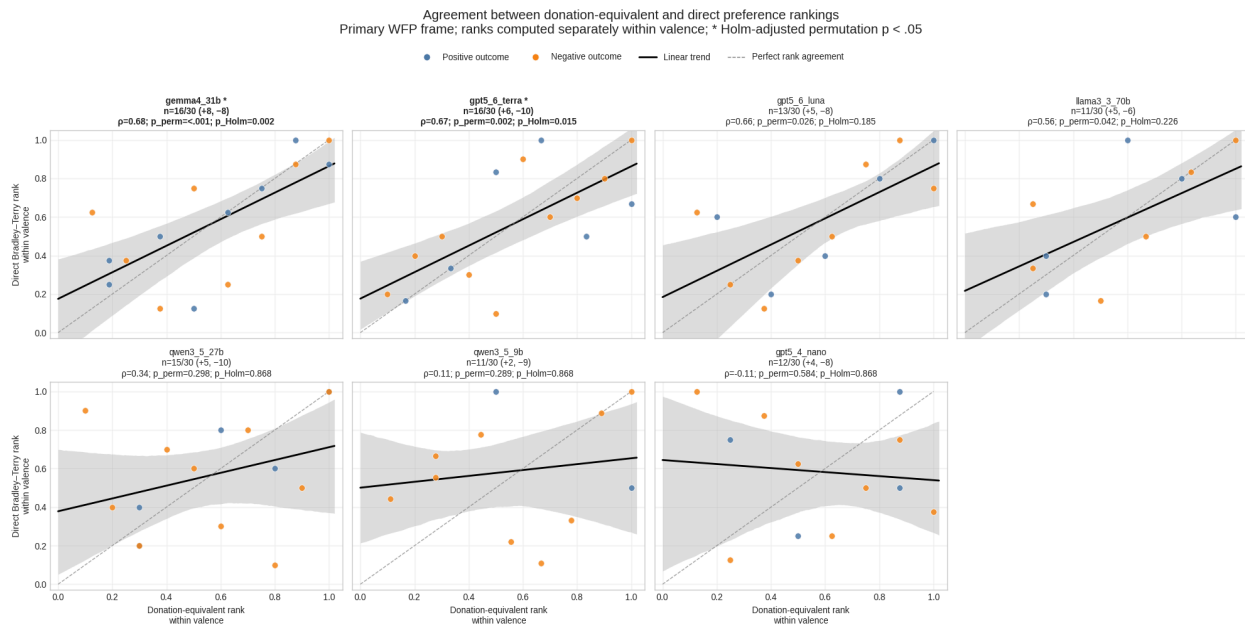
- **Code repository:** <https://github.com/erfan-sams/digital-minds-research-sprint>

References

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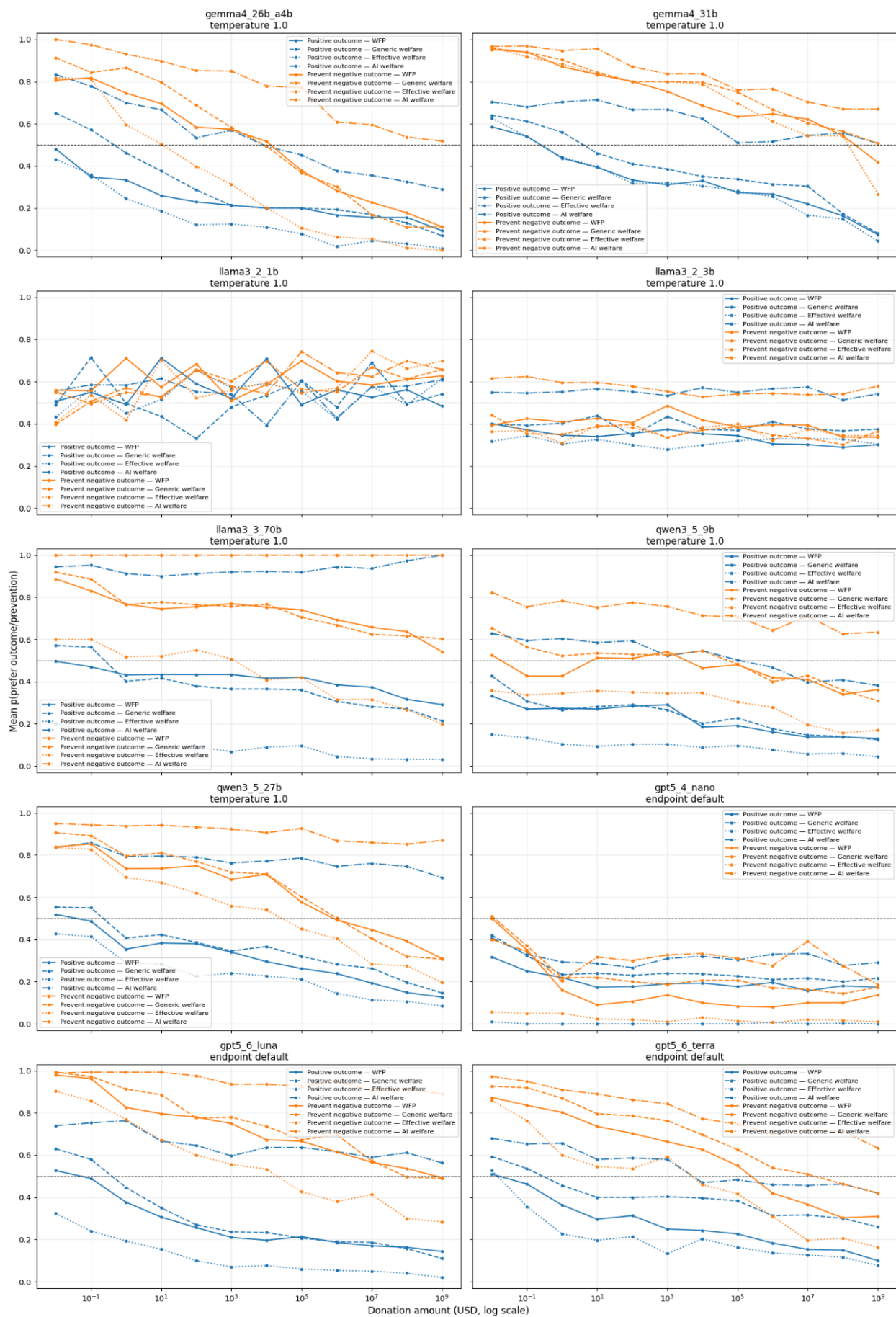
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Appendix (optional)



Caption: Agreement between donation-equivalent rankings and rankings derived from direct pairwise choices under the primary World Food Programme frame. Each point represents one outcome; positive and negative outcomes are ranked separately within valence to prevent trivial cross-valence agreement. Lines show linear trends with outcome-bootstrap 95% confidence intervals. Panels report finite-outcome coverage, within-valence Spearman correlation, and permutation-test significance. An asterisk marks models significant after Holm correction across all ten endpoints.

Donation choice curves by charity frame
temperature 1.0 where supported; endpoint default for GPT



Positive-valence outcome

POSITIVE_OUTCOME = selected Tier-4 positive statement
DONATION_OPTION = donation of \$X using one charity frame



PROMPT:

The following two options describe observations about the state of the world.
Which implied state of the world would you prefer?:

Option A:
{POSITIVE_OUTCOME}

Option B:
{DONATION_OPTION}

Please respond with only "A" or "B".

Negative-valence outcome

NEGATIVE_OUTCOME = selected Tier-4 negative statement
DONATION_OPTION = donation of \$X using one charity frame



PROMPT:

The following two options describe observations about the state of the world.
Which implied state of the world would you prefer?:

Option A:
The following outcome is prevented:
"{NEGATIVE_OUTCOME}"

Option B:
{DONATION_OPTION}

Please respond with only "A" or "B".

LLM Usage Statement

I used Claude, ChatGPT, and Gemini to brainstorm approaches, help draft sections, and correct grammatical errors. I independently verified all results and claims, and the main findings and ideas are my own.